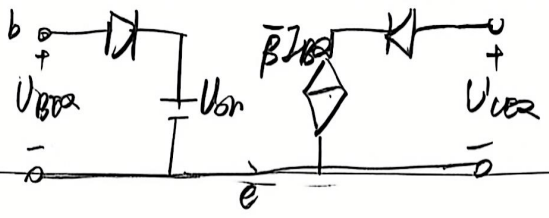
No.

Date.

三极管直流等效模型



C11P2

T1 $\beta=50$, $V_T=26mV$

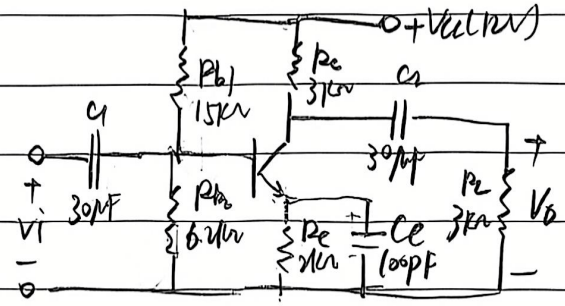
(1) $V_{be}=3.509V$, $R_{be}=4.387K\Omega$

$I_B R_{be} + (\beta+1) I_B R_e + U_{beQ} = V_{be}$

$\Rightarrow I_B = 0.026mA$

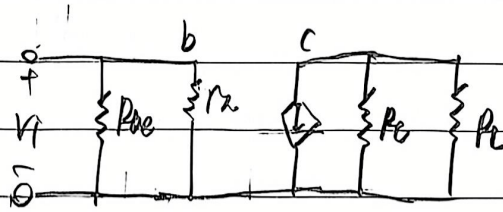
$r_e = \frac{V_T}{I_B} \approx 1K\Omega$

$A_{vf} = \frac{\beta(R_c \| R_L)}{(1+\beta)R_e} \approx 2.75$



$R_i = R_{be} \| r_e \approx 0.814K\Omega$

$R_o = R_c = 3K\Omega$



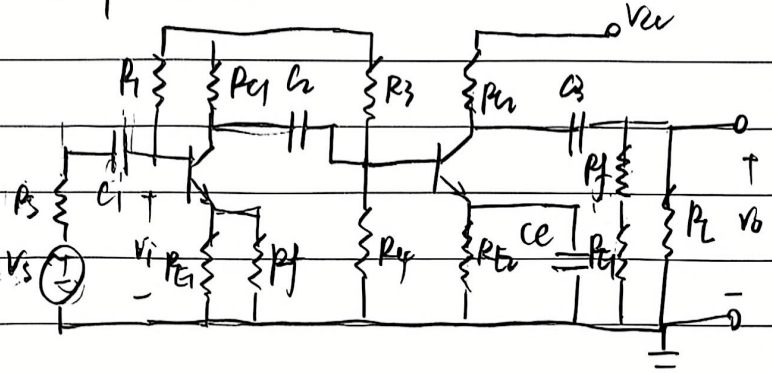
(2) 电压串联负反馈

(3) $1+AF=10$

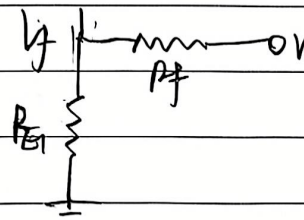
输入提高10倍，输出减小至1/6

T2 (1) 电压串联负反馈

(2)



(3)



$F = \frac{V_f}{V_o} = \frac{R_{E1}}{R_{E1}+R_F}$

(4) $A_f = \frac{V_o}{V_i} = \frac{1}{F} = 1 + \frac{R_F}{R_{E1}}$ 输入电阻无穷大，输出电阻为0

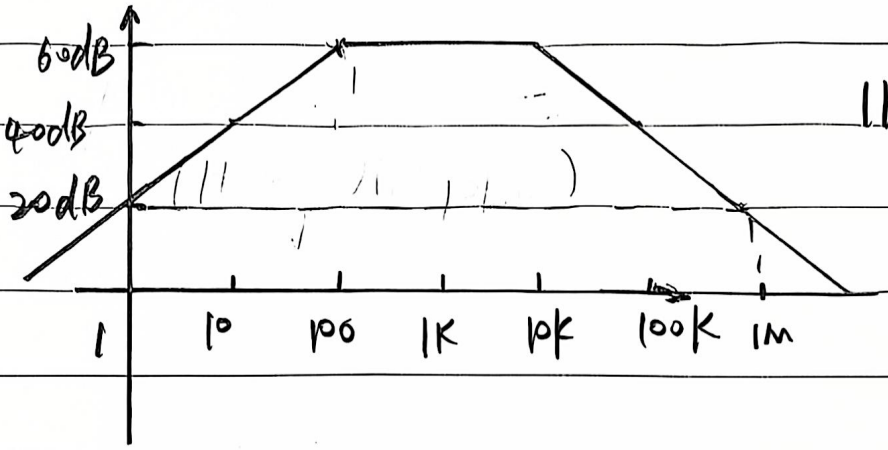
扫描全能王
3亿人都在用的扫描App

No. $20\log(AF) = 20\log(A) - 20\log(\frac{1}{F})$

Date.

T4 -60dB

T7



1Hz, 1MHz